Exploratory Data Analysis on Airbnb Listings Dataset

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv("AB_NYC_2019.csv")
```

In [4]:
```
df.head()
```

Out[4]:

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price	minimum_ni
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room	149	
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/apt	225	
2	3647	THE VILLAGE OF HARLEM....NEW YORK !	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.94190	Private room	150	
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/apt	89	
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/apt	80	

In [5]:
```
df.shape
```

Out[5]: (48895, 16)

In [6]:
```
df.columns
```

Out[6]:
```
Index(['id', 'name', 'host_id', 'host_name', 'neighbourhood_group',
       'neighbourhood', 'latitude', 'longitude', 'room_type', 'price',
       'minimum_nights', 'number_of_reviews', 'last_review',
       'reviews_per_month', 'calculated_host_listings_count',
       'availability_365'],
      dtype='object')
```

In [11]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 48895 entries, 0 to 48894
Data columns (total 13 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   id                              48895 non-null  int64
 1   host_id                         48895 non-null  int64
 2   neighbourhood_group             48895 non-null  object
 3   neighbourhood                   48895 non-null  object
 4   latitude                        48895 non-null  float64
 5   longitude                       48895 non-null  float64
 6   room_type                       48895 non-null  object
 7   price                           48895 non-null  int64
 8   minimum_nights                  48895 non-null  int64
 9   number_of_reviews               48895 non-null  int64
 10  reviews_per_month               48895 non-null  float64
 11  calculated_host_listings_count  48895 non-null  int64
 12  availability_365                48895 non-null  int64
dtypes: float64(3), int64(7), object(3)
memory usage: 4.8+ MB
```

In [12]:
```
df.describe()
```

Out[12]:

	id	host_id	latitude	longitude	price	minimum_nights	number_of_reviews	reviews_per_month	c
count	4.889500e+04	4.889500e+04	48895.000000	48895.000000	48895.000000	48895.000000	48895.000000	48895.000000	
mean	1.901714e+07	6.762001e+07	40.728949	-73.952170	152.720687	7.029962	23.274466	1.090910	
std	1.098311e+07	7.861097e+07	0.054530	0.046157	240.154170	20.510550	44.550582	1.597283	
min	2.539000e+03	2.438000e+03	40.499790	-74.244420	0.000000	1.000000	0.000000	0.000000	
25%	9.471945e+06	7.822033e+06	40.690100	-73.983070	69.000000	1.000000	1.000000	0.040000	
50%	1.967728e+07	3.079382e+07	40.723070	-73.955680	106.000000	3.000000	5.000000	0.370000	
75%	2.915218e+07	1.074344e+08	40.763115	-73.936275	175.000000	5.000000	24.000000	1.580000	
max	3.648724e+07	2.743213e+08	40.913060	-73.712990	10000.000000	1250.000000	629.000000	58.500000	

checking missing values

In [9]:

```
print(df.isnull().sum())
```

```
id                                    0
name                                 16
host_id                               0
host_name                            21
neighbourhood_group                   0
neighbourhood                         0
latitude                              0
longitude                             0
room_type                             0
price                                 0
minimum_nights                        0
number_of_reviews                     0
last_review                       10052
reviews_per_month                 10052
calculated_host_listings_count        0
availability_365                      0
dtype: int64
```

Handling Missing Values

In [10]:

```
df["reviews_per_month"] = df["reviews_per_month"].fillna(0)

df.drop(["last_review", "host_name", "name"], axis=1, inplace=True)
```

Check Duplicate Rows

In [13]:

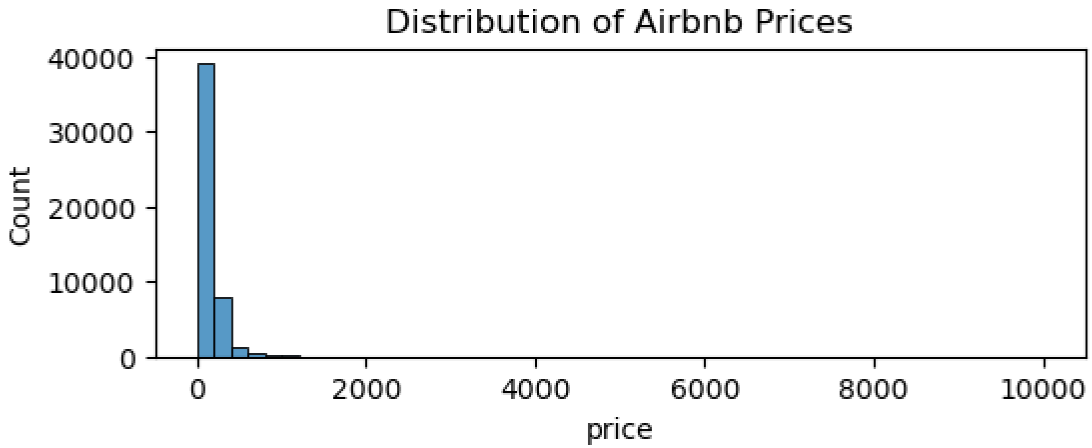
```
df.duplicated().sum()
```

Out[13]:

```
np.int64(0)
```

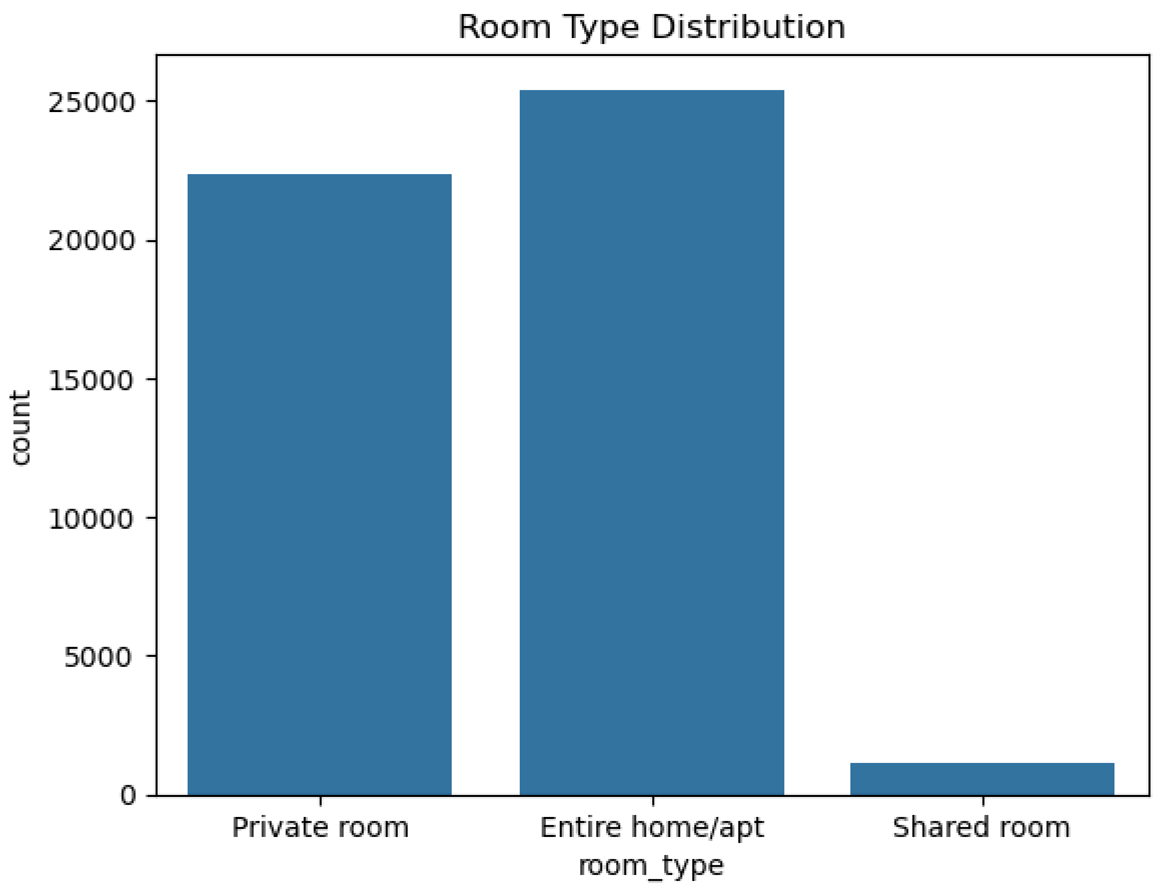
In [35]:

```
plt.figure(figsize=(6,2))
sns.histplot(df['price'], bins=50)
plt.title('Distribution of Airbnb Prices')
plt.show()
```



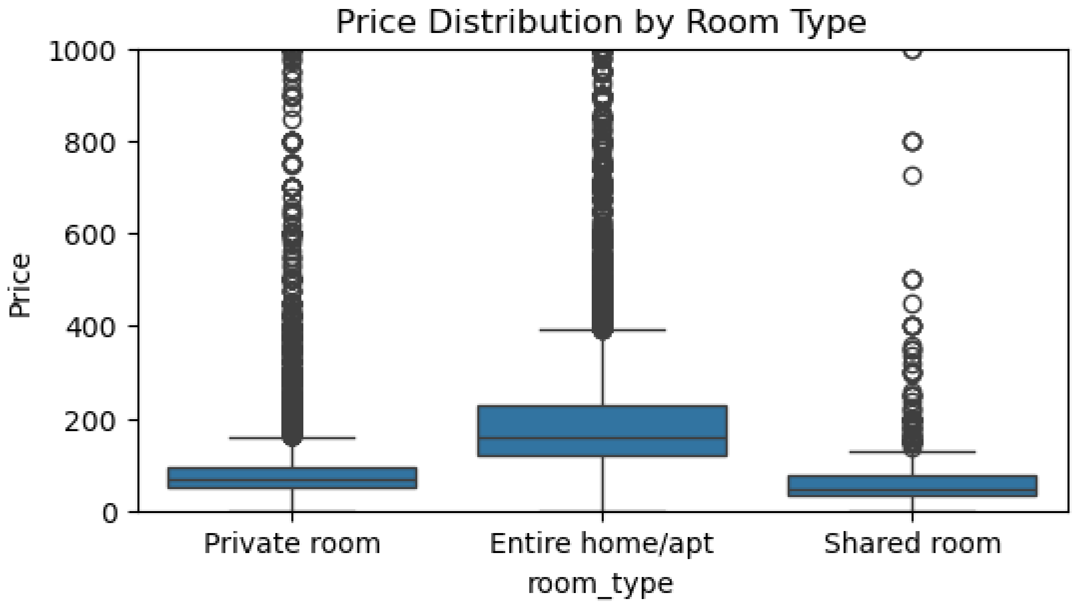
Most Airbnb listings are priced below 200 dollars, while a few listings have extremely high prices.

In [23]:
```
sns.countplot(x='room_type', data=df)
plt.title('Room Type Distribution')
plt.show()
```



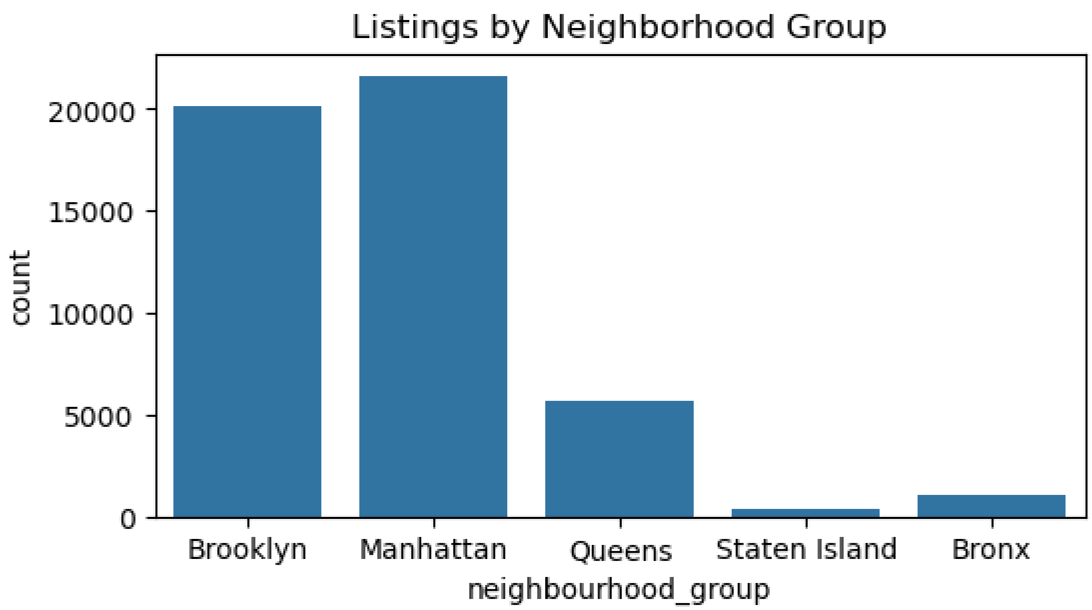
Entire homes/apartments are the most preferred room type among customers.

In [33]:
```
plt.figure(figsize=(6,3))
sns.boxplot(x='room_type', y='price', data=df)
plt.title('Price Distribution by Room Type')
plt.ylabel('Price')
plt.ylim(0, 1000)
plt.show()
```



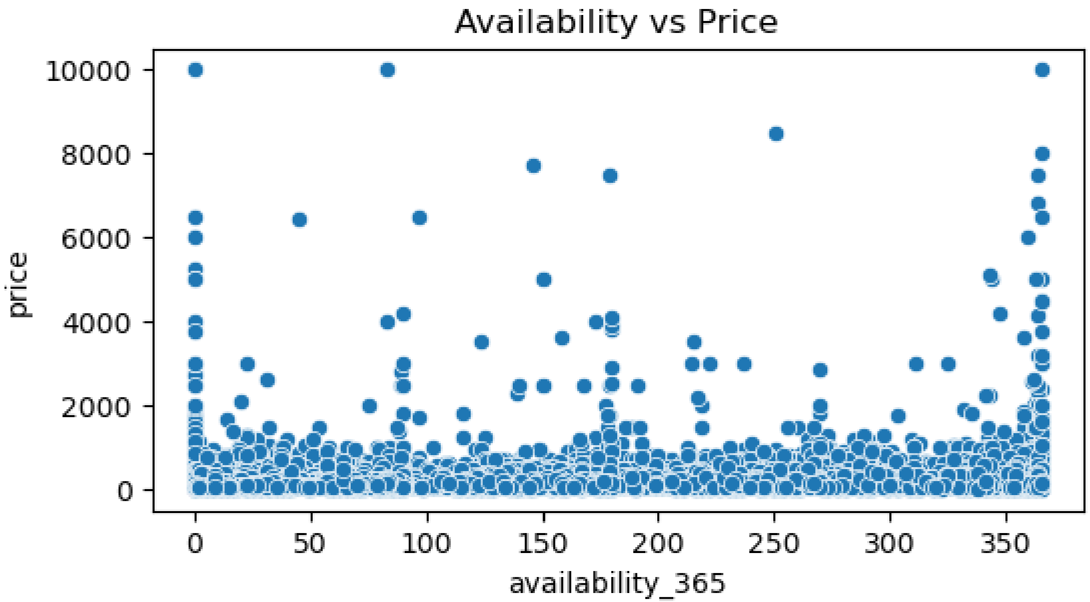
Entire homes/apartments generally have higher prices compared to private and shared rooms.

In [32]:
```
plt.figure(figsize=(6,3))
sns.countplot(x='neighbourhood_group', data=df)
plt.title('Listings by Neighborhood Group')
plt.show()
```



Manhattan and Brooklyn contain the highest number of Airbnb listings.

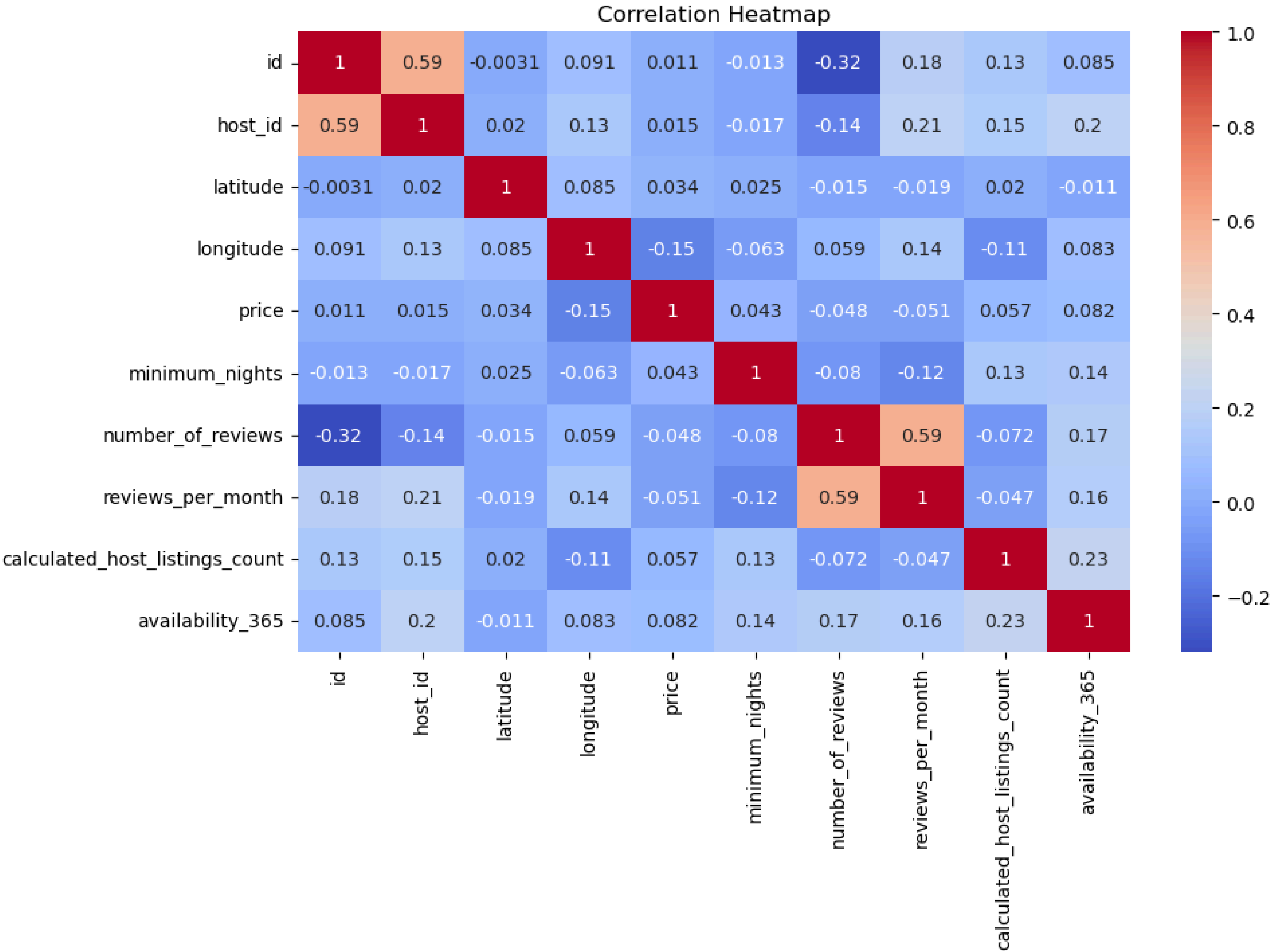
In [37]:
```
plt.figure(figsize=(6,3))
sns.scatterplot(x='availability_365', y='price', data=df)
plt.title('Availability vs Price')
plt.show()
```



Highly priced listings often have lower availability throughout the year.

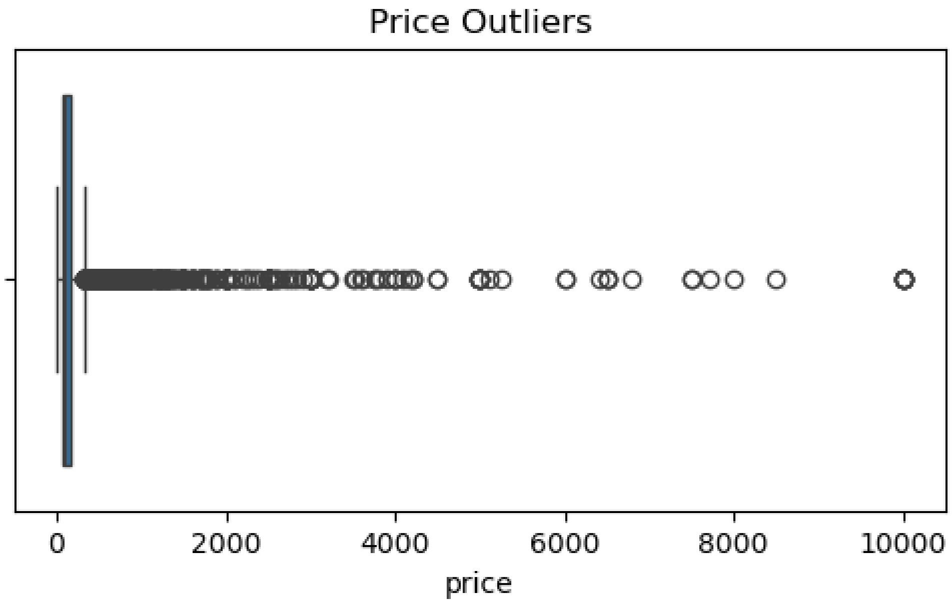
In [39]:
```
numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(10,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```


Correlation Heatmap
id
host_id
latitude
longitude
price
minimum_nights
number_of_reviews
reviews_per_month
calculated_host_listings_count
availability_365
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

Price has weak correlation with most numerical features, indicating that multiple factors influence pricing.

In [42]:
```
plt.figure(figsize=(6,3))
sns.boxplot(x=df['price'])
plt.title('Price Outliers')
plt.show()
```



Several extreme price outliers are present in the dataset.